



University
of Manitoba

Math prelims – Linear algebra

MATH 2740 – Mathematics of Data Science – Lecture 02

Julien Arino

`julien.arino@umanitoba.ca`

Department of Mathematics @ University of Manitoba

Fall 2026

The University of Manitoba campuses are located on original lands of Anishinaabeg, Ininew, Anisininew, Dakota and Dene peoples, and on the National Homeland of the Red River Métis. We respect the Treaties that were made on these territories, we acknowledge the harms and mistakes of the past, and we dedicate ourselves to move forward in partnership with Indigenous communities in a spirit of Reconciliation and collaboration.

In MATH 2740, we rely on notions you acquired in MATH 1210/1220/1300. We also use some material from first-year calculus

So let us (briefly) go over material in these courses

I also add (for some of you) a few things that will be handy and establish some terminology that we use throughout the course

Outline

Sets and logic

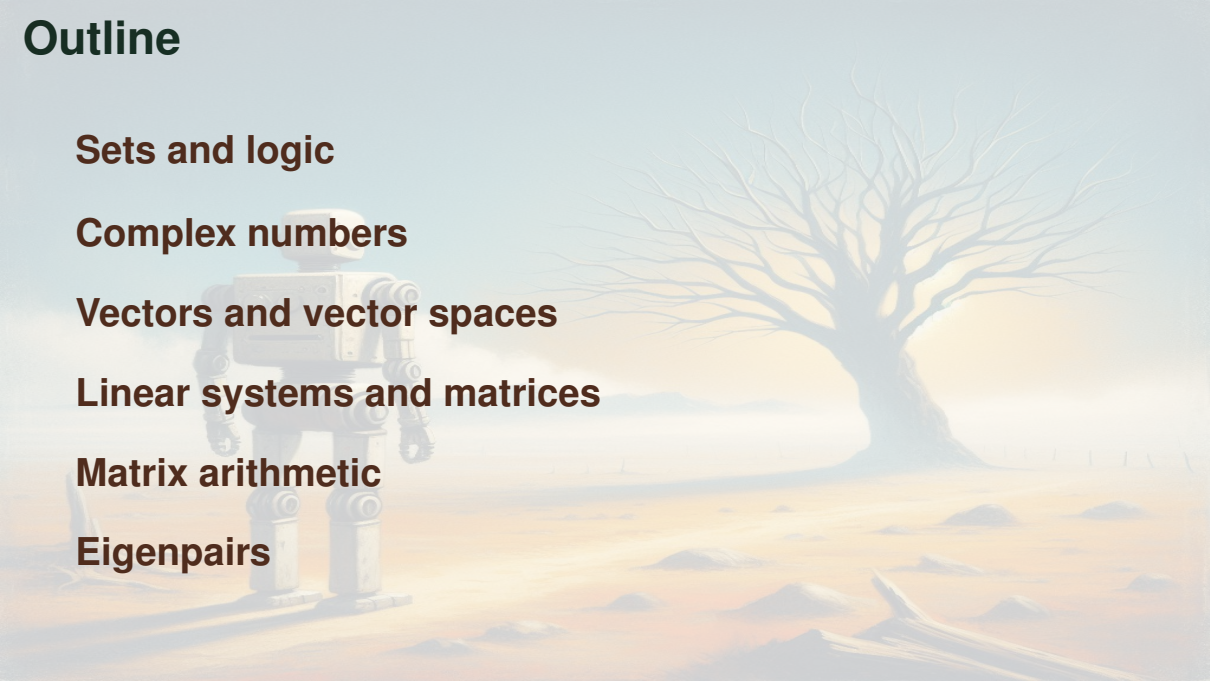
Complex numbers

Vectors and vector spaces

Linear systems and matrices

Matrix arithmetic

Eigenpairs



Linear algebra in a nutshell (MATH 1210/1220/1300)

Theorem 1

Let $A \in \mathcal{M}_n$. The following statements are equivalent (TFAE)

- 1. The matrix A is invertible*
- 2. $\forall \mathbf{b} \in \mathbb{F}^n$, $A\mathbf{x} = \mathbf{b}$ has a unique solution ($\mathbf{x} = A^{-1}\mathbf{b}$)*
- 3. The only solution to $A\mathbf{x} = \mathbf{0}$ is the trivial solution $\mathbf{x} = \mathbf{0}$*
- 4. $RREF(A) = \mathbb{I}_n$*
- 5. The matrix A is equal to a product of elementary matrices*
- 6. $\forall \mathbf{b} \in \mathbb{F}^n$, $A\mathbf{x} = \mathbf{b}$ has a solution*
- 7. There is a matrix $B \in \mathcal{M}_n$ such that $AB = \mathbb{I}_n$*
- 8. There is an invertible matrix $B \in \mathcal{M}_n$ such that $AB = \mathbb{I}_n$*
- 9. $\det(A) \neq 0$*
- 10. 0 is not an eigenvalue of A*

Sets and logic

Complex numbers

Vectors and vector spaces

Linear systems and matrices

Matrix arithmetic

Eigenpairs



Sets and elements

Definition 2 (Set)

A **set** X is a collection of **elements**

We write $x \in X$ or $x \notin X$ to indicate that the element x belongs to the set X or does not belong to the set X , respectively

Definition 3 (Subset)

Let X be a set. The set S is a **subset** of X , which is denoted $S \subset X$, if all its elements belong to X

Not used here but worth noting: we say S is a **proper subset** of X and write $S \subsetneq X$, if it is a subset of X and not equal to X

Quantifiers

A shorthand notation for “for all elements x belonging to X ” is $\forall x \in X$

For example, if $X = \mathbb{R}$, the *field* of real numbers, then $\forall x \in \mathbb{R}$ means “for all real numbers x ”

A shorthand notation for “there exists an element x in the set X ” is $\exists x \in X$

$\forall$ and $\exists$ are **quantifiers**

Intersection and union of sets

Let X and Y be two sets

Definition 4 (Intersection)

The intersection of X and Y , $X \cap Y$, is the set of elements that belong to X **and** to Y ,

$$X \cap Y = \{x : x \in X \text{ and } x \in Y\}$$

Definition 5 (Union)

The union of X and Y , $X \cup Y$, is the set of elements that belong to X **or** to Y ,

$$X \cup Y = \{x : x \in X \text{ or } x \in Y\}$$

In mathematics, or=and/or in common parlance. We also have an **exclusive or** (xor)

A teeny bit of logic

In a logical sense, a **proposition** is an assertion (or statement) whose truth value (true or false) can be asserted. For example, a theorem is a proposition that has been shown to be true. “The sky is blue” is also a proposition

Let A be a proposition. We generally write

A

to mean that A is true, and

not A

to mean that A is false. **not A** is the **contraposition** of A (or **not A** is the contrapositive of A)

A teeny bit of logic (cont.)

Let A, B be propositions. Then

- ▶ $A \Rightarrow B$ (read A implies B) means that whenever A is true, then so is B
- ▶ $A \Leftrightarrow B$, also denoted A if and only if B (A iff B for short), means that $A \Rightarrow B$ **and** $B \Rightarrow A$

We also say that A and B are **equivalent**

Let A and B be propositions. Then

$$(A \Rightarrow B) \Leftrightarrow (\text{not } B \Rightarrow \text{not } A)$$

Necessary or sufficient conditions

Suppose we want to establish whether a given statement P is true, depending on the truth value of a statement H . Then we say that

- ▶ H is a **necessary condition** if $P \Rightarrow H$
(It is necessary that H be true for P to be true; so whenever P is true, so is H)
- ▶ H is a **sufficient condition** if $H \Rightarrow P$
(It suffices for H to be true for P to also be true)
- ▶ H is a **necessary and sufficient condition** if $H \Leftrightarrow P$, i.e., H and P are equivalent

Playing with quantifiers

For the quantifiers $\forall$ (for all) and $\exists$ (there exists),

$\exists$ is the contrapositive of $\forall$

Therefore, for example, the contrapositive of

$$\forall x \in X, \exists y \in Y$$

is

$$\exists x \in X, \forall y \in Y$$

Sets and logic

Complex numbers

Vectors and vector spaces

Linear systems and matrices

Matrix arithmetic

Eigenpairs



Complex numbers

Definition 6 (Complex numbers)

A **complex number** is an ordered pair (a, b) , where $a, b \in \mathbb{R}$. Usually written $a + ib$ or $a + bi$, where $i^2 = -1$ (i.e., $i = \sqrt{-1}$)

The set of all complex numbers is denoted $\mathbb{C}$,

$$\mathbb{C} = \{a + ib : a, b \in \mathbb{R}\}$$

Definition 7 (Addition and multiplication on $\mathbb{C}$)

Letting $a + ib$ and $c + id \in \mathbb{C}$, addition on $\mathbb{C}$ is defined by

$$(a + ib) + (c + id) = (a + c) + i(b + d)$$

and multiplication on $\mathbb{C}$ is defined by

$$(a + ib)(c + id) = (ac - bd) + i(ad + bc)$$

Latter is easy to obtain using regular multiplication and $i^2 = -1$

Properties

$$\forall \alpha, \beta, \gamma \in \mathbb{C},$$

$$\alpha + \beta = \beta + \alpha \text{ and } \alpha\beta = \beta\alpha$$

[commutativity]

$$(\alpha + \beta) + \gamma = \alpha + (\beta + \gamma) \text{ and } (\alpha\beta)\gamma = \alpha(\beta\gamma)$$

[associativity]

$$\gamma + 0 = \gamma \text{ and } \gamma 1 = \gamma$$

[identities]

$$\forall \alpha \in \mathbb{C}, \exists \beta \in \mathbb{C} \text{ unique s.t. } \alpha + \beta = 0$$

[additive inverse]

$$\forall \alpha \neq 0 \in \mathbb{C}, \exists \beta \in \mathbb{C} \text{ unique s.t. } \alpha\beta = 1$$

[multiplicative inverse]

$$\gamma(\alpha + \beta) = \gamma\alpha + \gamma\beta$$

[distributivity]

Additive & multiplicative inverse, subtraction, division

Definition 8

Let $\alpha, \beta \in \mathbb{C}$

- ▶ $-\alpha$ is the **additive inverse** of α , i.e., the unique number in $\mathbb{C}$ s.t. $\alpha + (-\alpha) = 0$

- ▶ **Subtraction** on $\mathbb{C}$:

$$\beta - \alpha = \beta + (-\alpha)$$

- ▶ For $\alpha \neq 0$, $1/\alpha$ is the **multiplicative inverse** of α , i.e., the unique number in $\mathbb{C}$ s.t.

$$\alpha(1/\alpha) = 1$$

- ▶ **Division** on $\mathbb{C}$:

$$\beta/\alpha = \beta(1/\alpha)$$

Definition 9 (Real and imaginary parts)

Let $z = a + ib$. Then $\operatorname{Re} z = a$ is **real part** and $\operatorname{Im} z = b$ is **imaginary part** of z

If ambiguous, write $\operatorname{Re}(z)$ and $\operatorname{Im}(z)$

Definition 10 (Conjugate and Modulus)

Let $z = a + ib \in \mathbb{C}$. Then

- ▶ **Complex conjugate** of z is

$$\bar{z} = a - ib$$

- ▶ **Modulus** (or **absolute value**) of z is

$$|z| = \sqrt{a^2 + b^2} \geq 0$$

Properties of complex numbers

Let $w, z \in \mathbb{C}$, then

- ▶ $z + \bar{z} = 2\operatorname{Re} z$
- ▶ $z - \bar{z} = 2i\operatorname{Im} z$
- ▶ $z\bar{z} = |z|^2$
- ▶ $\overline{w + z} = \bar{w} + \bar{z}$ and $\overline{wz} = \bar{w}\bar{z}$
- ▶ $\overline{\bar{z}} = z$
- ▶ $|\operatorname{Re} z| \leq |z|$ and $|\operatorname{Im} z| \leq |z|$
- ▶ $|\bar{z}| = |z|$
- ▶ $|wz| = |w| |z|$
- ▶ $|w + z| \leq |w| + |z|$

[triangle inequality]

Solving quadratic equations

Consider the polynomial

$$P(x) = a_0 + a_1x + a_2x^2$$

where $x, a_0, a_1, a_2 \in \mathbb{R}$. Letting

$$\Delta = a_1^2 - 4a_0a_2$$

you know that if $\Delta > 0$, then

$$P(x) = 0$$

has two distinct *real* solutions,

$$x_1 = \frac{-a_1 - \sqrt{\Delta}}{2a_2} \quad \text{and} \quad x_2 = \frac{-a_1 + \sqrt{\Delta}}{2a_2}$$

if $\Delta = 0$, then there is a (multiplicity 2) unique *real* solution

$$x_1 = \frac{-a_1}{2a_2}$$

while if $\Delta < 0$, there is no solution

Solving quadratic equations with complex numbers

Consider the polynomial

$$P(x) = a_0 + a_1x + a_2x^2$$

where $x, a_0, a_1, a_2 \in \mathbb{R}$. If instead of seeking $x \in \mathbb{R}$, we seek $x \in \mathbb{C}$, then the situation is the same, except when $\Delta < 0$

In the latter case, note that

$$\sqrt{\Delta} = \sqrt{(-1)(-\Delta)} = \sqrt{-1}\sqrt{-\Delta} = i\sqrt{-\Delta}$$

Since $\Delta < 0$, $-\Delta > 0$ and the square root is the usual one

Solving quadratic equations with complex numbers

To summarize, consider the polynomial

$$P(x) = a_0 + a_1x + a_2x^2$$

where $x, a_0, a_1, a_2 \in \mathbb{R}$. Letting

$$\Delta = a_1^2 - 4a_0a_2$$

Then

$$P(x) = 0$$

has two solutions,

$$x_{1,2} = \frac{-a_1 \pm \sqrt{\Delta}}{2a_2}$$

where, if $\Delta < 0$, $x_1, x_2 \in \mathbb{C}$ and take the form

$$x_{1,2} = \frac{-a_1 \pm i\sqrt{-\Delta}}{2a_2}$$

Why this matters

Recall (we will come back to this later) that to find the *eigenvalues* of the matrix

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

we seek λ solutions to $\det(A - \lambda\mathbb{I}) = 0$, i.e., λ solutions to

$$|A - \lambda\mathbb{I}| = \begin{vmatrix} a_{11} - \lambda & a_{12} \\ a_{21} & a_{22} - \lambda \end{vmatrix} = (a_{11} - \lambda)(a_{22} - \lambda) - a_{12}a_{21} = 0$$

i.e., λ solutions to

$$\lambda^2 - (a_{11} + a_{22})\lambda + a_{11}a_{22} - a_{12}a_{21} = 0$$

Why this matters (cont.)

Let

$$P(\lambda) = \lambda^2 - (a_{11} + a_{22})\lambda + a_{11}a_{22} - a_{12}a_{21}$$

From previous discussion, letting

$$\begin{aligned}\Delta &= (a_{11} + a_{22})^2 - 4(a_{11}a_{22} - a_{12}a_{21}) \\ &= a_{11}^2 + a_{22}^2 + 2a_{11}a_{22} - 4a_{11}a_{22} + 4a_{12}a_{21} \\ &= a_{11}^2 + a_{22}^2 - 2a_{11}a_{22} + 4a_{12}a_{21} \\ &= (a_{11} - a_{22})^2 + 4a_{12}a_{21}\end{aligned}$$

we have two (potentially equal) solutions to $P(\lambda) = 0$

$$x_{1,2} = \frac{a_{11} + a_{22} \pm \sqrt{\Delta}}{2}$$

that are complex if $\Delta < 0$

Example: $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$

Sets and logic

Complex numbers

Vectors and vector spaces

Linear systems and matrices

Matrix arithmetic

Eigenpairs



Vectors

A **vector** $\mathbf{v}$ is an ordered n -tuple of real or complex numbers

Denote $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$ (real or complex numbers). For $v_1, \dots, v_n \in \mathbb{F}$,

$$\mathbf{v} = (v_1, \dots, v_n) \in \mathbb{F}^n$$

is a vector. $v_1, \dots, v_n$ are the **components** of $\mathbf{v}$

If unambiguous, we write v . Otherwise, $\mathbf{v}$ or $\vec{v}$

Vector space

Definition 11 (Vector space)

A **vector space** over $\mathbb{F}$ is a set V together with two binary operations, **vector addition**, denoted $+$, and **scalar multiplication**, that satisfy the relations:

1. $\forall \mathbf{u}, \mathbf{v}, \mathbf{w} \in V, \mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$
2. $\forall \mathbf{v}, \mathbf{w} \in V, \mathbf{v} + \mathbf{w} = \mathbf{w} + \mathbf{v}$
3. $\exists \mathbf{0} \in V$, the zero vector, such that $\mathbf{v} + \mathbf{0} = \mathbf{v}$ for all $\mathbf{v} \in V$
4. $\forall \mathbf{v} \in V$, there exists an element $\mathbf{w} \in V$, the additive inverse of $\mathbf{v}$, such that $\mathbf{v} + \mathbf{w} = \mathbf{0}$
5. $\forall \alpha \in \mathbb{R}$ and $\forall \mathbf{v}, \mathbf{w} \in V, \alpha(\mathbf{v} + \mathbf{w}) = \alpha\mathbf{v} + \alpha\mathbf{w}$
6. $\forall \alpha, \beta \in \mathbb{R}$ and $\forall \mathbf{v} \in V, (\alpha + \beta)\mathbf{v} = \alpha\mathbf{v} + \beta\mathbf{v}$
7. $\forall \alpha, \beta \in \mathbb{R}$ and $\forall \mathbf{v} \in V, \alpha(\beta\mathbf{v}) = (\alpha\beta)\mathbf{v}$
8. $\forall \mathbf{v} \in V, 1\mathbf{v} = \mathbf{v}$

Norms

Definition 12 (Norm)

Let V be a vector space over $\mathbb{F}$, and $\mathbf{v} \in V$ be a vector. The **norm** of $\mathbf{v}$, denoted $\|\mathbf{v}\|$, is a function from V to $\mathbb{R}_+$ that has the following properties:

1. For all $\mathbf{v} \in V$, $\|\mathbf{v}\| \geq 0$ with $\|\mathbf{v}\| = 0$ iff $\mathbf{v} = \mathbf{0}$
2. For all $\alpha \in \mathbb{F}$ and all $\mathbf{v} \in V$, $\|\alpha\mathbf{v}\| = |\alpha| \|\mathbf{v}\|$
3. For all $\mathbf{u}, \mathbf{v} \in V$, $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$

Let V be a vector space (for example, $\mathbb{R}^2$ or $\mathbb{R}^3$)

The **zero element** (or **zero vector**) is the vector $\mathbf{0} = (0, \dots, 0)$

The **additive inverse** of $\mathbf{v} = (v_1, \dots, v_n)$ is $-\mathbf{v} = (-v_1, \dots, -v_n)$

For $\mathbf{v} = (v_1, \dots, v_n) \in V$, the length (or Euclidean norm) of $\mathbf{v}$ is the **scalar**

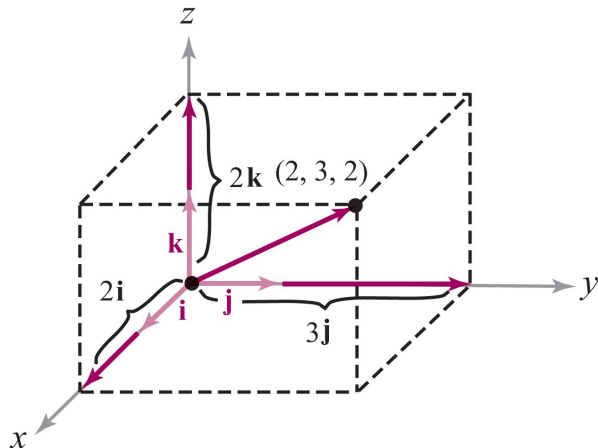
$$\|\mathbf{v}\| = \sqrt{v_1^2 + \dots + v_n^2}$$

To **normalize** the vector $\mathbf{v}$ consists in considering $\tilde{\mathbf{v}} = \mathbf{v}/\|\mathbf{v}\|$, i.e., the vector in the same direction as $\mathbf{v}$ that has unit length

Standard basis vectors

Vectors $\mathbf{i} = (1, 0, 0)$, $\mathbf{j} = (0, 1, 0)$ and $\mathbf{k} = (0, 0, 1)$ are the **standard basis vectors** of $\mathbb{R}^3$. A vector $\mathbf{v} = (v_1, v_2, v_3)$ can then be written

$$\mathbf{v} = v_1 \mathbf{i} + v_2 \mathbf{j} + v_3 \mathbf{k}$$



For $V(\mathbb{R}^n)$, the standard basis vectors are usually denoted $\mathbf{e}_1, \dots, \mathbf{e}_n$, with

$$\mathbf{e}_k = (\underbrace{0, \dots, 0}_{k-1}, 1, \underbrace{0, \dots, 0}_{n-k+1})$$

Dot product

Definition 13 (Dot product)

Let $\mathbf{a} = (a_1, \dots, a_n) \in \mathbb{R}^n$, $\mathbf{b} = (b_1, \dots, b_n) \in \mathbb{R}^n$. The **dot product** of $\mathbf{a}$ and $\mathbf{b}$ is the **scalar**

$$\mathbf{a} \bullet \mathbf{b} = \sum_{i=1}^n a_i b_i = a_1 b_1 + \dots + a_n b_n$$

The dot product is a special case of **inner product**

Properties of the dot product

Theorem 14

For $\mathbf{a}, \mathbf{b}, \mathbf{c} \in \mathbb{R}^n$ and $\alpha \in \mathbb{R}$,

▶ $\mathbf{a} \bullet \mathbf{a} = \|\mathbf{a}\|^2$

(so $\mathbf{a} \bullet \mathbf{a} \geq 0$, with $\mathbf{a} \bullet \mathbf{a} = 0$ iff $\mathbf{a} = \mathbf{0}$)

▶ $\mathbf{a} \bullet \mathbf{b} = \mathbf{b} \bullet \mathbf{a}$

($\bullet$ is commutative)

▶ $\mathbf{a} \bullet (\mathbf{b} + \mathbf{c}) = \mathbf{a} \bullet \mathbf{b} + \mathbf{a} \bullet \mathbf{c}$

($\bullet$ distributive over $+$)

▶ $(\alpha \mathbf{a}) \bullet \mathbf{b} = \alpha(\mathbf{a} \bullet \mathbf{b}) = \mathbf{a} \bullet (\alpha \mathbf{b})$

▶ $\mathbf{0} \bullet \mathbf{a} = 0$

Some results stemming from the dot product

Theorem 15

If θ is the angle between the vectors $\mathbf{a}$ and $\mathbf{b}$, then

$$\mathbf{a} \bullet \mathbf{b} = \|\mathbf{a}\| \|\mathbf{b}\| \cos \theta$$

Corollary 16 (Cauchy-Schwarz inequality)

For any two vectors $\mathbf{a}$ and $\mathbf{b}$, we have

$$|\mathbf{a} \bullet \mathbf{b}| \leq \|\mathbf{a}\| \|\mathbf{b}\|$$

with equality if and only if $\mathbf{a}$ is a scalar multiple of $\mathbf{b}$, or one of them is $\mathbf{0}$.

Theorem 17

$\mathbf{a}$ and $\mathbf{b}$ are orthogonal if and only if $\mathbf{a} \bullet \mathbf{b} = 0$.

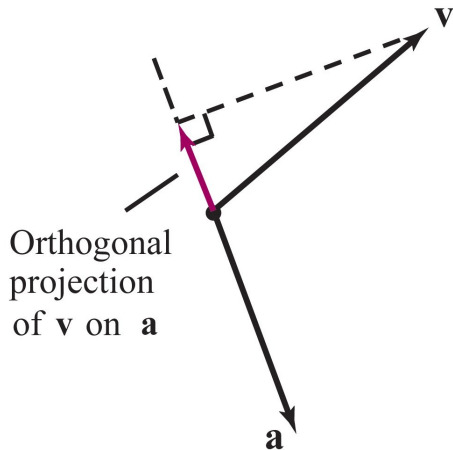
Scalar and vector projections

Scalar projection of $\mathbf{v}$ onto $\mathbf{a}$ (or component of $\mathbf{v}$ along $\mathbf{a}$):

$$\text{comp}_{\mathbf{a}} \mathbf{v} = \frac{\mathbf{a} \bullet \mathbf{v}}{\|\mathbf{a}\|}$$

Vector (or orthogonal) projection of $\mathbf{v}$ onto $\mathbf{a}$:

$$\text{proj}_{\mathbf{a}} \mathbf{v} = \left(\frac{\mathbf{a} \bullet \mathbf{v}}{\|\mathbf{a}\|} \right) \frac{\mathbf{a}}{\|\mathbf{a}\|} = \frac{\mathbf{a} \bullet \mathbf{v}}{\|\mathbf{a}\|^2} \mathbf{a}$$



A robot stands in a vast, arid desert landscape. The robot is a boxy, mechanical figure with a rectangular head, a torso with circular details, and jointed limbs. It stands on the left side of the frame, facing right. In the background, a large, leafless tree with a complex, branching structure stands prominently. Beyond the tree, rolling hills or mountains are visible under a pale, hazy sky. The ground is sandy and dotted with small, dark rocks. The overall atmosphere is one of isolation and quiet observation.

Sets and logic

Complex numbers

Vectors and vector spaces

Linear systems and matrices

Matrix arithmetic

Eigenpairs

Linear systems

Definition 18 (Linear system)

A **linear system** of m equations in n unknowns takes the form

$$\begin{array}{ccccccccc} a_{11}x_1 & + & a_{12}x_2 & + & \cdots & + & a_{1n}x_n & = & b_1 \\ a_{21}x_1 & + & a_{22}x_2 & + & \cdots & + & a_{2n}x_n & = & b_2 \\ \vdots & & \vdots & & \vdots & & \vdots & & \vdots \\ a_{m1}x_1 & + & a_{m2}x_2 & + & \cdots & + & a_{mn}x_n & = & b_n \end{array} \quad (1)$$

The a_{ij} , x_j and b_j could be in $\mathbb{R}$ or $\mathbb{C}$, although here we typically assume they are in $\mathbb{R}$

The aim is to find $x_1, x_2, \dots, x_n$ that satisfy all equations simultaneously

Theorem 19 (Nature of solutions to a linear system)

A linear system can have

- ▶ *no solution*
- ▶ *a unique solution*
- ▶ *infinitely many solutions*

Operations on linear systems

You learned to manipulate linear systems using

- ▶ Gaussian elimination
- ▶ Gauss-Jordan elimination

with the aim to put the system in **row echelon form** (REF) or **reduced row echelon form** (RREF)

Matrices and linear systems

Writing

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \quad \text{and} \quad \mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix}$$

where A is an $m \times n$ **matrix**, $\mathbf{x}$ and $\mathbf{b}$ are n (column) **vectors** (or $n \times 1$ matrices), then the linear system in the previous slide takes the form

$$A\mathbf{x} = \mathbf{b}$$

Notation for vectors

We usually assume vectors are column vectors and thus write, e.g.,

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = (x_1, x_2, \dots, x_n)^T$$

Here, T is the **transpose operator** (more on this soon)

Consider the system

$$A\mathbf{x} = \mathbf{b}$$

If $\mathbf{b} = \mathbf{0}$, the system is **homogeneous** and always has the solution $\mathbf{x} = \mathbf{0}$ and so the “no solution” option in Theorem 19 goes away

Sets and logic

Complex numbers

Vectors and vector spaces

Linear systems and matrices

Matrix arithmetic

Eigenpairs



Definition 20 (Matrix)

An m -by- n or $m \times n$ matrix is a rectangular array of elements of $\mathbb{R}$ or $\mathbb{C}$ with m rows and n columns,

$$A = [a_{ij}] = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix}$$

We always list indices as “row,column”

We denote $\mathcal{M}_{mn}(\mathbb{F})$ or $\mathbb{F}^{mn}$ the set of $m \times n$ matrices with entries in $\mathbb{F} = \{\mathbb{R}, \mathbb{C}\}$. Often, we omit $\mathbb{F}$ in $\mathcal{M}_{mn}$ if the nature of $\mathbb{F}$ is not important

When $m = n$, we usually write $\mathcal{M}_n$

Basic matrix arithmetic

Let $A \in \mathcal{M}_{mn}$, $B \in \mathcal{M}_{mn}$ be matrices (of the same size) and $c \in \mathbb{F} = \{\mathbb{R}, \mathbb{C}\}$ be a scalar

► **Scalar multiplication**

$$cA = [ca_{ij}]$$

► **Addition**

$$A + B = [a_{ij} + b_{ij}]$$

► **Subtraction** (addition of $-B = (-1)B$ to A)

$$A - B = A + (-1)B = [a_{ij} + (-1)b_{ij}] = [a_{ij} - b_{ij}]$$

► **Transposition** of A gives a matrix $A^T \in \mathcal{M}_{nm}$ with

$$A^T = [a_{ji}], \quad j = 1, \dots, n, \quad i = 1, \dots, m$$

Matrix multiplication

The (matrix) **product** of A and B , AB , requires the “inner dimensions” to match, i.e., the number of columns in A must equal the number of rows in B

Suppose that is the case, i.e., let $A \in \mathcal{M}_{mn}$, $B \in \mathcal{M}_{np}$. Then the i, j entry in $C := AB$ takes the form

$$c_{ij} = \sum_{k=1}^n a_{ik} b_{kj}$$

Recall that the matrix product is not commutative, i.e., in general, $AB \neq BA$ (when both those products are defined, i.e., when $A, B \in \mathcal{M}_n$)

Special matrices

Definition 21 (Zero and identity matrices)

The **zero** matrix is the matrix 0_{mn} whose entries are all zero. The **identity** matrix is a square $n \times n$ matrix $\mathbb{I}_n$ with all entries on the main diagonal equal to one and all off diagonal entries equal to zero

Definition 22 (Symmetric matrix)

A square matrix $A \in \mathcal{M}_n$ is **symmetric** if $\forall i, j = 1, \dots, n, a_{ij} = a_{ji}$. In other words, $A \in \mathcal{M}_n$ is symmetric if $A = A^T$

Making symmetric matrices

Theorem 23

1. *If $A \in \mathcal{M}_n$, then $A + A^T$ is symmetric*
2. *If $A \in \mathcal{M}_{mn}$, then $AA^T \in \mathcal{M}_m$ and $A^T A \in \mathcal{M}_n$ are symmetric*

Proof of Theorem 23

X symmetric $\iff X = X^T$, so use $X =$ the matrix whose symmetric property you want to check

1. True if $A + A^T = (A + A^T)^T$. We have

$$(A + A^T)^T = A^T + (A^T)^T = A^T + A = A + A^T$$

2. AA^T symmetric if $AA^T = (AA^T)^T$. We have

$$(AA^T)^T = (A^T)^T A^T = AA^T$$

$A^T A$ works similarly

Determinants

Definition 24 (Determinant)

Let $A \in \mathcal{M}_n$ with $n \geq 2$. The **determinant** of A is the *scalar*

$$\det(A) = |A| = \sum_{j=1}^n a_{ij} C_{ij}$$

where $C_{ij} = (-1)^{i+j} \det(A_{ij})$ is the (i, j) -**cofactor** of A and A_{ij} is the submatrix of A from which the i th row and j th column have been removed

This is a cofactor expansion along the i th row

This is a recursive formula: it gives result in terms of $n-1$ $\mathcal{M}_{n-1}$ matrices, to which it must in turn be applied, all the way down to

$$\det \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} = a_{11} a_{22} - a_{12} a_{21}$$

Two special matrices and their determinants

Definition 25

$A \in \mathcal{M}_n$ is **upper triangular** if $a_{ij} = 0$ when $i > j$, **lower triangular** if $a_{ij} = 0$ when $j > i$, **triangular** if it is *either* upper or lower triangular and **diagonal** if it is *both* upper and lower triangular

When A diagonal, we often write $A = \text{diag}(a_{11}, a_{22}, \dots, a_{nn})$

Theorem 26

Let $A \in \mathcal{M}_n$ be triangular or diagonal. Then

$$\det(A) = \prod_{i=1}^n a_{ii} = a_{11} a_{22} \cdots a_{nn}$$

Inversion/Singularity

Definition 27 (Matrix inverse)

$A \in \mathcal{M}_n$ is **invertible** (or **nonsingular**) if $\exists A^{-1} \in \mathcal{M}_n$ s.t.

$$AA^{-1} = A^{-1}A = \mathbb{I}$$

A^{-1} is the **inverse** of A . If A^{-1} does not exist, A is **singular**

Theorem 28

Let $A \in \mathcal{M}_n$, $\mathbf{x}, \mathbf{b} \in \mathbb{F}^n$. Then

- ▶ A invertible $\iff \det(A) \neq 0$
- ▶ If A invertible, A^{-1} is unique
- ▶ If A invertible, then $A\mathbf{x} = \mathbf{b}$ has the unique solution $\mathbf{x} = A^{-1}\mathbf{b}$

Revisiting matrix arithmetic

With addition, subtraction, scalar multiplication, multiplication, transposition and inversion, you can perform arithmetic on matrices essentially as on scalar, if you bear in mind a few rules

- ▶ The sizes have to be compatible
- ▶ The order is important since matrix multiplication is not commutative
- ▶ Transposition and inversion change the order of products:

$$(AB)^T = B^T A^T \text{ and } (AB)^{-1} = B^{-1} A^{-1}$$

Sets and logic

Complex numbers

Vectors and vector spaces

Linear systems and matrices

Matrix arithmetic

Eigenpairs



Eigenvalues / Eigenvectors / Eigenpairs

Definition 29

Let $A \in \mathcal{M}_n$. A vector $\mathbf{x} \in \mathbb{F}^n$ such that $\mathbf{x} \neq \mathbf{0}$ is an **eigenvector** of A if $\exists \lambda \in \mathbb{F}$ called an **eigenvalue**, s.t.

$$A\mathbf{x} = \lambda\mathbf{x}$$

A couple $(\lambda, \mathbf{x})$ with $\mathbf{x} \neq \mathbf{0}$ s.t. $A\mathbf{x} = \lambda\mathbf{x}$ is an **eigenpair**

If $(\lambda, \mathbf{x})$ eigenpair, then for $c \neq 0$, $(\lambda, c\mathbf{x})$ also eigenpair since $A(c\mathbf{x}) = cA\mathbf{x} = c\lambda\mathbf{x}$ and dividing both sides by c .